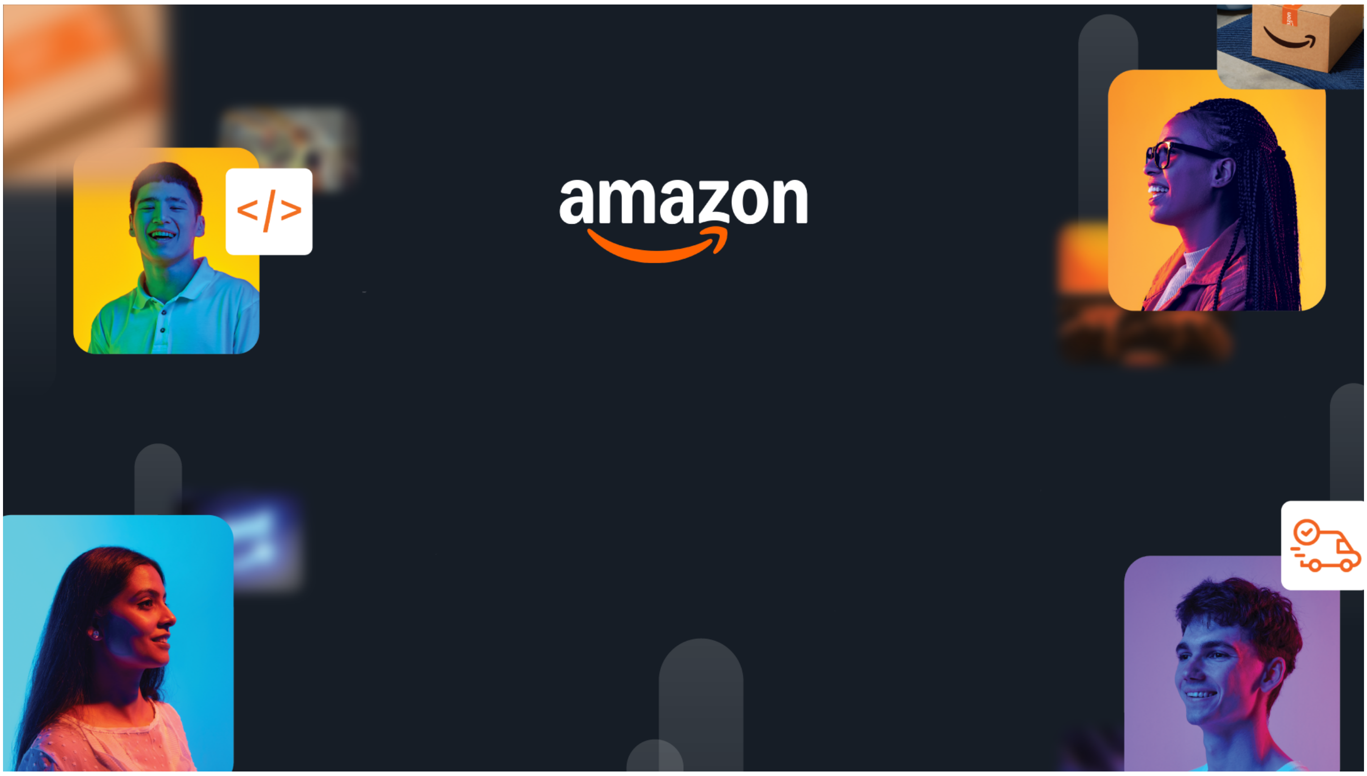

HackOn with Amazon

A Universe of Opportunity

48-Hour Hackathon I Solution Document

Team Name	Codyssey
Hackathon Theme	Amazon Now - Reimagining Urgent Shopping
Date	15/06/2026

Team Members

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1. Problem Statement & Relevance

The Problem

- Quick commerce has solved delivery — a 10-minute promise — but not deciding.
- Across 300M+ online grocery shoppers, users still spend 8–12 minutes per order manually searching each item, comparing variants, checking stock, and improvising substitutes when something is unavailable.
- The friction that used to live in the warehouse now lives in the search bar — and no system yet takes a plain need and hands back a cart ready for checkout.

Why It Matters

- **Who:** 300M+ online grocery shoppers globally (50M+ in India) waste 8–12 minutes per session assembling carts for outcomes they already know — across every quick-commerce platform (Blinkit, Zepto, Instamart, Amazon Fresh).
- **How large:** This "deciding tax" sits on top of a $50B+ market that perfected 10-minute delivery but left the 10-minute decision bottleneck completely unsolved.
- **Cost of inaction:** High cart abandonment (an industry-wide ~70% problem in online retail) means billions in lost GMV every year — every abandoned cart is lost revenue, every manual substitution erodes trust, and every friction-filled session risks permanent churn.

Theme Alignment

Theme connection — NowCart addresses all three opportunity areas with one coherent product:

- **Frictionless Shopping —** invisible out-of-stock substitution removes comparison paralysis; the user never hits a dead end.
- **Shopping by Intent —** five front doors (voice, budget, photo, recipe link, and zero-input prediction) replace the search box entirely.
- **Predictive & Confident —** Zero Door predicts your restock before you ask, and every pick carries a confidence score with one-line reasoning.

Unique angle — Every competitor optimizes the search box; NowCart removes it. Users don't want products, they want outcomes — so we work backwards from a budget, a dish photo, or even a YouTube cooking video turned into a shoppable cart, with confidence and substitution built in as a core layer, not a popup.

What Makes This Novel

Core novelty — Comparison Collapse. Instead of showing six variants of ghee and asking the user to choose, the AI picks one, defends it with a confidence score and a one-line reason ("organic, best price per litre, you've bought this brand before — 92%"), and only asks the human when overall cart confidence drops too low. This flips the model from "here are your options" to "here's my answer — challenge me if you disagree."

What's architecturally new. Voice, photo, recipe link, and budget all converge into a single intent, then flow through a shared-state reasoning pipeline where each step builds on the last — substitution reads the match results, confidence reads the substitution changes. That makes the logic composable and auditable, not one fragile prompt. No grocery platform has a reasoning pipeline — they have search indexes.

2. Customer & Solution

Target Customer

The Time-Starved Outcome Shopper — urban, 22–40, opens a quick-commerce app already knowing the outcome (biryani for four, a restock before guests arrive, dinner for two on ₹500) but not the line-item list to get there. They cook a few times a week, treat browsing as a cost rather than a feature, and abandon the moment they hit choice paralysis or an out-of-stock item.

What they need — not a better search box, but an interface that takes an intent ("biryani for four") and hands back a ready-to-checkout cart, with smart substitutions baked in so one missing item never breaks the flow.

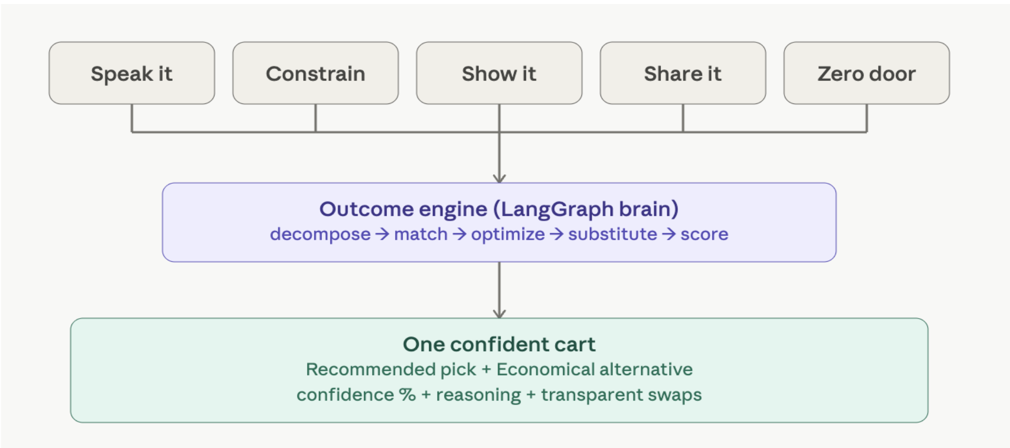
Secondary users — voice-first elders delegating the family shop, accessibility users locked out of text-search interfaces, and budget-bound students who shop by constraint rather than by item.

How We Solve It

NowCart turns a plain need into a checkout-ready cart. Four ways:

1. **Multiple front doors, one engine —** every input feeds the same brain:
    - **Speak It —** say or type a moment ("biryani for 4") and it becomes a cart.
    - **Constrain It —** give a budget, and the engine works backwards to fit it.
    - **Show It —** snap a photo of a dish, and get its ingredients as a cart.
    - **Share It —** drop a recipe link, YouTube URL, or text, and it becomes a cart.
    - **Zero Door —** give nothing; the system predicts your restock before you ask.
2. **One confident pick, not a results page —** instead of 20 lookalikes, we show one, with a reason and confidence score attached, plus a cheaper Economical swap. The same logic powers search: best match leads, alternatives rank below.
3. **Resilient, pantry-aware, predictive —** out-of-stock items are swapped automatically with a transparent audit trail, so the cart never breaks. The engine skips what you already have, and Zero Door builds a restock cart from your purchase patterns — shifting from reactive to anticipatory.
4. **SOS Mode —** one tap for real emergencies ("guests in 30 minutes"): an in-stock-only kit with quantities and delivery ETA on a single screen, ready for two-tap checkout — collapsing the whole decision into one glance when time is shortest.

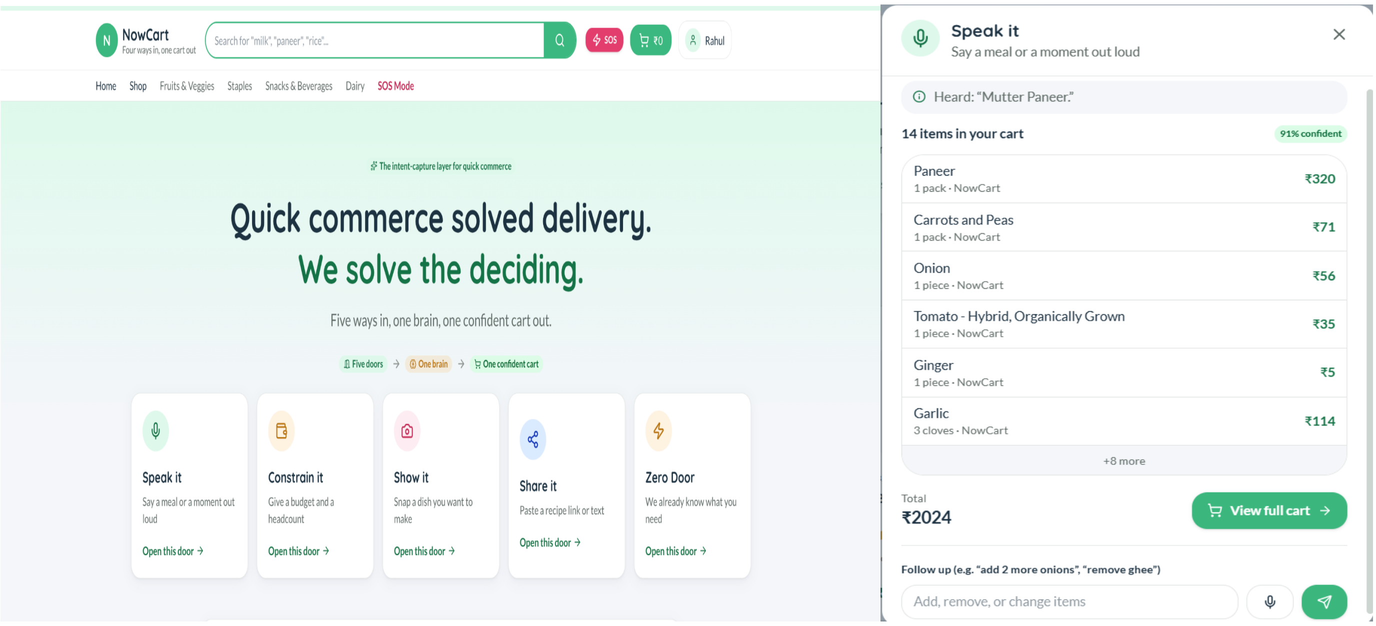
User Workflow

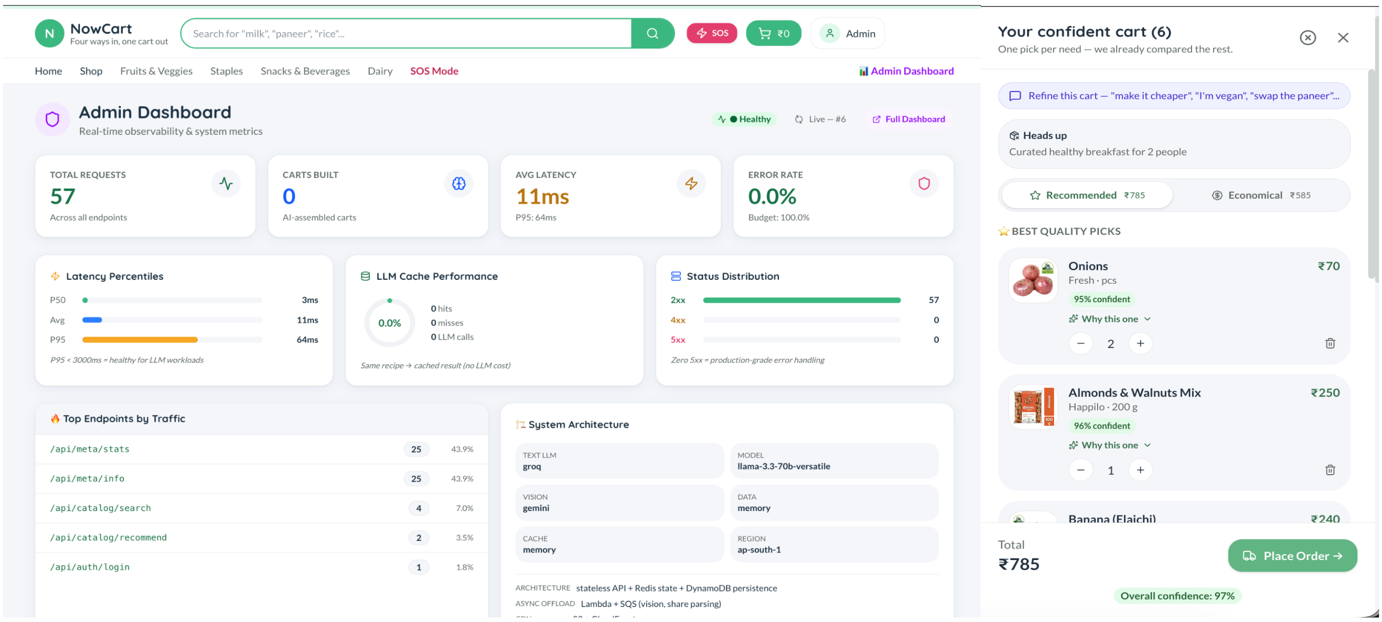


3-step flow in words:

1. **Express:** open any front door: speak a meal, set a budget, snap a photo, paste a link, or just let it predict.
2. **Engine:** the pipeline decomposes the intent, matches it against 9,534 products, optimizes the picks, resolves out-of-stock items, and scores its confidence.
3. **Confirm:** one cart appears with confidence scores, a reasoning trail, and transparent substitutions — then the user refines it conversationally ("make it cheaper," "I'm vegan") or checks out in two taps.

Working Prototype



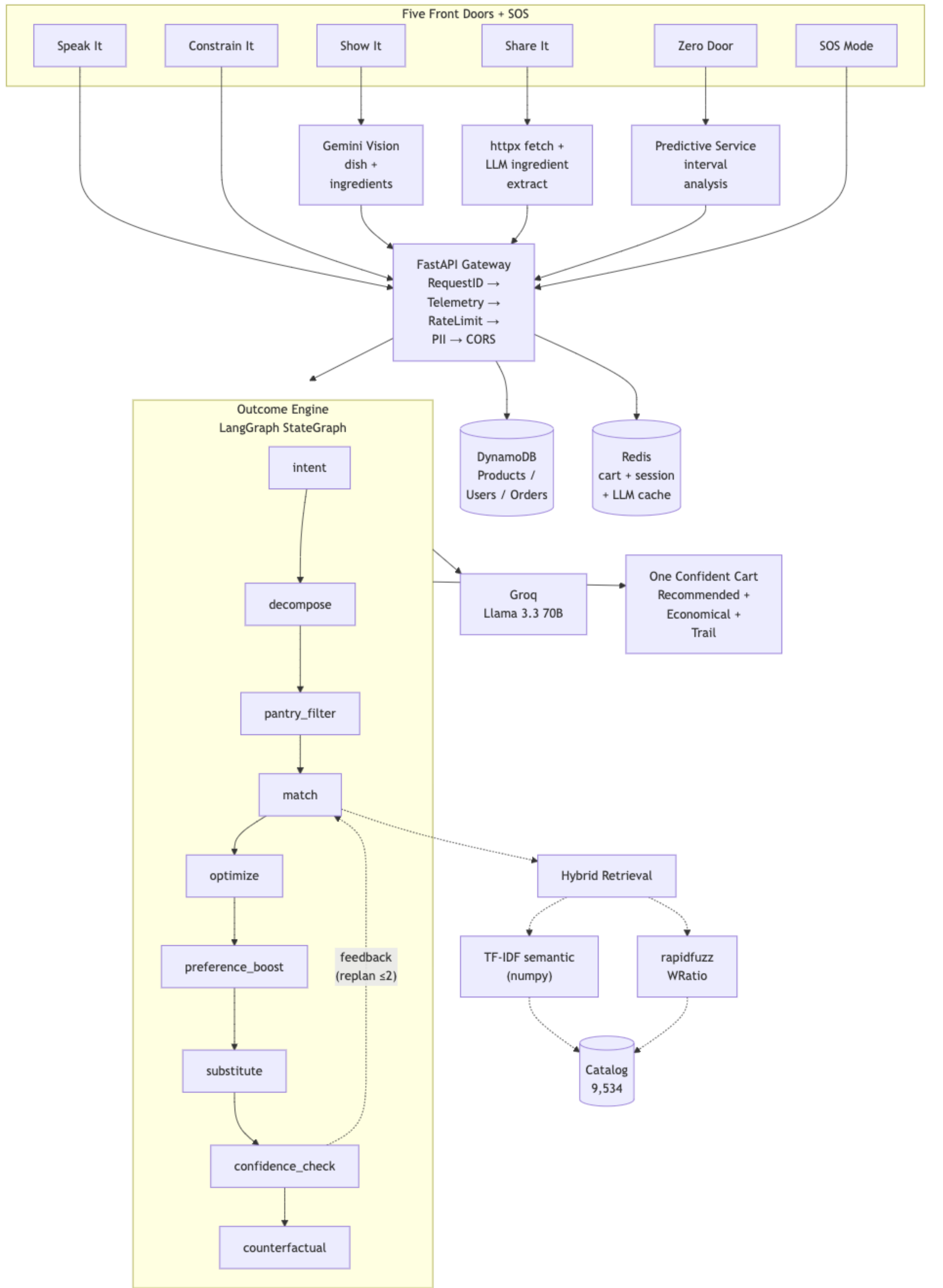


Demo: [Demo Video / Live App / Source Code]

Deployed on AWS and fully functional. For the complete experience (pantry awareness, predictions, order history), sign in as rahul@gmail.com — any password works.

3. Tech Architecture & Scaling

Architecture



Tech Stack

Layer	Technology	Why
Frontend	React 19 + Vite + TailwindCSS	Fast build/HMR and type-safe API calls via auto-generated OpenAPI types; zero-runtime CSS keeps the five front-door panels snappy
Backend	FastAPI + Pydantic 2 (fully async)	Native async I/O, automatic OpenAPI schema generation, and typed request/response validation on every DTO
AI / ML	LangGraph + Groq (Llama 3.3 70B) / Gemini 2.0 Flash / Bedrock (Claude 3 Haiku) + pure-NumPy TF-IDF & rapidfuzz	LangGraph runs the 10-node Outcome Engine as a composable multi-agent DAG; provider abstraction swaps LLMs with one env var; hybrid TF-IDF + fuzzy retrieval matches needs to products with zero model download
Database	DynamoDB (PAY_PER_REQUEST)	Auto-scales the 9,534-product catalog plus users and orders with no capacity planning; GSI for category queries; order history feeds the predictive Zero Door
Infra	EC2 + Nginx + S3 + CloudFront + Redis (Lambda + SQS designed)	React build served globally via CloudFront from S3; FastAPI on EC2 behind Nginx with co-located Redis for cart/session/LLM caching; SQS + Lambda wired for async LLM offload as traffic grows

Key Algorithms & Complexity

- **LangGraph Multi-Agent Pipeline** — A 10-node reasoning graph with shared typed state and a self-correcting replan loop (capped at 2 passes). Each node builds on the last, so the logic is composable and debuggable instead of one brittle mega-prompt. Complexity: linear in needs × catalog size.
- **Hybrid Retrieval (TF-IDF + Fuzzy)** — A pure-NumPy character n-gram index with a Hindi/English synonym table narrows 9,534 products by meaning, then rapidfuzz scores the shortlist by name. No model download, no GPU, sub-second index build. Catches both "malai → cream" and typos. Complexity: index build is linear; per-query is near-constant on the shortlist.
- **Confidence-Sorted Budget Fill** — Items are ranked by confidence and kept greedily until the budget runs out, so the most trustworthy picks always survive the cut. Complexity: one sort plus one pass.
- **Predictive Restock (Zero Door)** — Measures how often you rebuy each item, projects when you'll run out, and scores certainty from how regular the pattern is. Pure statistics — no training, no cold start. Complexity: linear in products × order history.
- **LLM Response Caching** — Each prompt is hashed to a 1-hour Redis entry, so 100 identical queries cost 1 model call and 99 sub-millisecond cache hits — cutting both spend and ~800 ms of latency per repeat.

Scaling Strategy

- **Stateless API:** Cart state lives in Redis, not in the server process, so any instance can serve any request. 100x: an Auto Scaling Group behind a load balancer adds instances on demand. 1000x: multi-region groups with latency-based routing send users to their nearest region.
- **Database:** no capacity planning. DynamoDB on-demand auto-scales reads and writes natively, with no sharding or provisioning. 1000x: Global Tables replicate across regions for fast local reads worldwide.
- **LLM offloading (scaling path):** Slow AI calls (vision ~3s, recipe parsing ~2s) move to Lambda via SQS, auto-scaling to 1000 parallel executions. This decouples API response time from AI processing time entirely — the API stays fast no matter how heavy the model work gets.

- **Multi-layer caching:** In-memory hot catalog data, Redis for carts and LLM responses (hashed, 1-hour TTL), and DynamoDB DAX for catalog reads at scale. Common intents like "biryani for four" become pure cache hits — instant and free.
- **Frontend:** already global. S3 + CloudFront serves the app from edge locations worldwide today, with no change needed to handle millions of users.
- **Provider abstraction:** Switching from Groq to Amazon Bedrock is a single environment variable. At scale, Bedrock runs inside our AWS network with managed throughput and no third-party rate limits.

4. Future Vision

Where This Goes

NowCart becomes the universal "need → done" layer on top of any fulfillment network. The engine is category-agnostic — it works anywhere a person has a goal and needs products assembled, not just groceries.

Roadmap

Horizon	Milestone	Impact
0-3 mo	Grocery intent-capture live in one Indian metro — voice, budget, photo, and share all functional	1K active users, 3x faster cart assembly vs. manual, under 15% cart abandonment
3-6 mo	Pharmacy SOS (OTC health kits), multi-language voice (Hindi, Tamil, Bengali), predictive restock at scale	10K users, language barrier removed for 500M+ non-English speakers
6-12 mo	B2B restocking (restaurants, offices), event and travel kits, white-label API integration with grocery platforms	20k+ users, licensing revenue per cart built, horizontal "need → done" layer proven

Multi-Segment Expansion

NowCart's engine is domain-agnostic: it applies to any market where people know the outcome they want but not the exact products to buy. The expansion path moves from consumer essentials to high-frequency commercial use:

- **Grocery (today)** → "Biryani for 4" becomes a full ingredient cart in seconds — proving the core thesis.
- Pharmacy & Health → "Child has a fever" → OTC medicine, hydration, and a comfort kit. The highest-urgency, highest-trust market — same pipeline, health catalog.
- **Creator & Social Commerce** → any cooking video, reel, or blog becomes a shoppable cart. We already turn a YouTube link into ingredients today; the path is partnering with creators and platforms so recipe content is one-tap buyable. The content already exists — we make it purchasable.
- **B2B Restocking** → "Restock my restaurant for the week" or "office pantry for 50" → bulk carts with quantity scaling, budget limits, and recurring auto-orders. Higher order values, predictable recurring revenue.
- Travel & Event Kits → "Camping trip for 6" or "party for 20 kids" → a single cart spanning food, supplies, and gear across categories.

The Path: consumer (grocery) → urgent care (pharmacy) → content-driven (creator commerce) → commercial (B2B) → cross-category (events). Each step reuses the identical engine — adding a vertical is a catalog and prompt change, not an architecture rebuild. Any industry where humans have a goal but face product overload is addressable.

Value Impact

- **Users reached.** Approx 300M+ online grocery shoppers globally (50M+ in India), with multi-segment expansion adding 200M+ across pharmacy, B2B, and travel. Critically, voice-first and photo-first input unlocks an estimated 100M+ users excluded by text-only interfaces — the elderly, visually impaired, and non-English speakers.
- **Time saved.** ~10 minutes per session × 3 sessions/week ≈ 26 hours returned per user per year. At 1M users, that's 26M hours of human time given back annually — an entire friction layer removed from daily life.
- **Efficiency gains.** Confident carts with pre-resolved out-of-stock items directly attack cart abandonment (an industry-wide ~70% problem in online retail). LLM response caching removes up to 99% of redundant model calls at scale, so cost stays flat as repeat queries grow.
- **Revenue potential.** A white-label "cart-built" pricing model: at a few cents per cart and 10M carts/month, that's a six-figure monthly licensing revenue stream — on top of faster carts driving higher order frequency and platform GMV for partners.

Links: *GitHub* / *Demo Video* / *Live App*